BuckAI Observatory

Center of Excellence in Applied AI for Earth and Planetary Observation

Mission. BuckAI Observatory advances OSU's leadership in artificial intelligence (AI) by integrating cutting-edge AI algorithms with remote-sensing science. We foster a collaborative, interdisciplinary, and inclusive community that develops, applies, and disseminates AI-enabled Earth and Planetary Observation solutions, while providing shared resources, expert support, and educational opportunities.

How the Mission Is Achieved

1. **Driving research excellence**
   - Develop and deploy AI tools for multi-modal, time-stacked satellite, airborne, and in-situ data.
   - Build and/or use Foundation Models for geoscience and related disciplines.
   - Catalyze proposal teams to pursue diverse research funding through federal grants, industry partnerships, technology commercialization, and/or philanthropic support.
2. **Providing shared cyberinfrastructure and expertise**
   - Manage dedicated GPU nodes and large storage volumes on ASC's *Unity* cluster.
   - Maintain college-level data-engineering pipelines, code repositories, and documentation.
   - Offer proposal-development, compliance, and "red-team" reviews in partnership with ERIK.
3. **Cultivating interdisciplinary collaboration**
   - Connect investigators across multiple Arts and Sciences departments, and beyond.
   - Host boot camps, seminars, and a recurring AI + Remote-Sensing symposium.
4. **Educating the next generation**
   - Coordinate data-centric and AI coursework across departments.
   - Provide hands-on training spaces, undergraduate labeling internships, and GRAs.
   - Embed responsible-AI principles in curricula and outreach.
5. **Engaging society**
   - Communicate scientific findings and societal applications of AI-enabled Earth observation to policymakers, stakeholders, and the public.
   - Support open-source tools and reproducible research that benefit the broader community.